Algebra 1
Unit 1
Lesson 2 Homework

Name Answer Key ____________
Date____________________
Period___________________

Given the function $r(x) = -\frac{2}{3}x + 4$, find each value.

1. $r(-6) = -\frac{2}{3}(-6) + 4$

$= \frac{12}{3} + 4 = 4 + 4 = \boxed{8}$

2. $r\left(\frac{1}{2}\right) = -\frac{2}{3}\left(\frac{1}{2}\right) + 4$

$= -\frac{1}{3} + 4 = \boxed{\frac{11}{3}}$

3. For what value of x is $r(x) = -4$?

$3\left[-4 = -\frac{2}{3}x + 4\right]3$

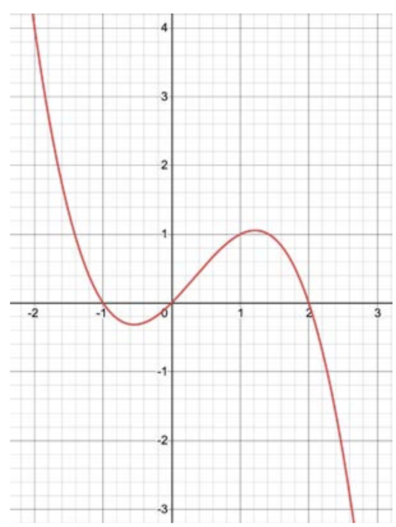
$-12 = -2x + 12$

$\underline{-12 \qquad\qquad -12}$

$-24 = -2x$

$\boxed{x = 12}$

Use the graph of $y = f(x)$ to find each value.



4. $f(-2) = 4$

5. $f(1) = 1$

6. For what value of x is $f(x) = 0$?

$x = 0$

Use the table of values representing $\boldsymbol{m(x)}$ to find each value.

$\boldsymbol{x}$	$\boldsymbol{m(x)}$
-6	4.5
8	2
0	-3.2
-24	-13
7	-9
3	0

7. $m(8) =$ 2

8. $m(-24) =$ -13

9. What is the y-intercept of the function in the table?
$(0, -3.2)$

10. Given $d(x) = -2|x^2 - 3.6|$, find the value for $x = 5.5$.
$d(5.5) = -2|(5.5)^2 - 3.6|$
$= -2|30.25 - 3.6|$
$= -2|26.65|$
$\boxed{d(5.5) = -53.3}$